

PHYS 225

Week 4, Lecture 4:
Derivation of the Lorentz Transformations



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reminders

- Homework 3 due at 11:59 pm this Thursday, 2/12
- Office Hours **tomorrow 2/11 4:00-7:00pm, Loomis 276**
and **Thursday 2/12 5:00-7:00pm, Loomis 279**
- Don't forget to turn in your **1-minute papers** by 6 pm today!

Week 4, Lecture 4

Derivation of the Lorentz transformations

Today we will derive the algebraic relationships between the coordinates in S and S' , using **only** the two postulates of relativity.

This relationship is so special it is given its own name: the **Lorentz transformations**. These are the relativistic versions of the Galilean transformations from Week 1.

One Minute Paper: Name at least one way the Galilean transformations of Newtonian mechanics must be modified to be consistent with relativity.

Week 4, Lecture 4

Derivation of the Lorentz transformations

To the blackboard!

$S(t, x, y, z)$

$S'(t', x', y', z')$

L.4 |

- (I) The laws of physics are the same in S and S'
(II) The speed of light is the same in S and S'

- Synchronizing the origins of S and S' :

At $t=t'=0$ $x=y=z=0$ and $x'=y'=z'=0$.

- Let S' move with velocity v in x direction $\Rightarrow$
 $\Rightarrow x'=0$ corresponds to $x=vt$ in S
- Consider 2 beams of light in the $+x$ and $-x$ directions.
- for $+x$: $x=ct$ in S and $x'=ct'$ in S' $\Leftrightarrow$ (II) $\Rightarrow$
 $\Rightarrow x-ct=0 \Leftrightarrow x'-ct'=0 \Rightarrow$ (I) requires coord. transformations to be linear
 $\Rightarrow x'-ct' = \lambda(x-ct)$ (1) λ is some number.
- for $-x$: $x=-ct$, $x'=-ct' \Rightarrow$
 $\Rightarrow x'+ct' = \mu(x+ct)$ (2)
- let's define $a = \frac{\lambda+\mu}{2}$, $b = \frac{\lambda-\mu}{2}$.
- (1) + (2) $\rightarrow 2x' = (\lambda+\mu)x - (\lambda-\mu)ct \Rightarrow x' = ax - bct$ (3)
- (1) - (2) $\rightarrow -2ct' = (\lambda-\mu)x - (\lambda+\mu)ct \Rightarrow ct' = -bx + act$ (4)
- By definition (*): $x'=0 \Rightarrow x=vt$ plugging this in to (3) gives: $0 = avt - bct \Rightarrow b = \frac{av}{c}$ (5)

- Plugging (5) in to (4), (3) gives:

$$\boxed{\begin{aligned} x' &= a(x - vt) & (6) \\ ct' &= a(ct - \frac{vx}{c}) & (7) \end{aligned}}$$

- ⑥ tells us how S' sees events in S . (4) 2
According to postulate (I), the inverse transformation (telling us how S sees events in S') must be obtained by switching the sign of V .
 S and S' are on equal footing $\Rightarrow$
 $\Rightarrow$ if S' moves with velocity V w.r.t. S , then
 S moves with velocity $-V$ w.r.t. S' . $\Rightarrow$
 $\Rightarrow \boxed{x = a(x' + Vt')} \quad (8)$

- On the other hand we have ⑥ and ⑦

⑦ gives $\boxed{t = \frac{t'}{a} + \frac{Vx}{c^2}}$ $\xrightarrow{\text{plug into ⑥}}$ $x' = ax - Vt' - \frac{aV^2}{c^2}x$

$\Rightarrow \boxed{x' + Vt' = a(1 - \frac{V^2}{c^2})x} \quad (9)$ (10)

• compare ⑧ and ⑨ $\Rightarrow a = \frac{1}{a(1 - \frac{V^2}{c^2})} \Rightarrow a = \gamma = \frac{1}{\sqrt{1 - \frac{V^2}{c^2}}}$

• Note $\gamma(V) = \gamma(-V)$

- Note we took positive root in (10) because as $V \rightarrow 0$ directions of x and x' in S and S' must ~~not~~ be the same $\Rightarrow$ we need $a \rightarrow 1$.

- Thus we have obtained Lorentz transformations

$\boxed{\begin{matrix} x' = \gamma(x - Vt) & a \\ t' = \gamma(t - \frac{Vx}{c^2}) & b \end{matrix}} \quad (11) \quad \text{or} \quad \boxed{\begin{matrix} x = \gamma(x' + Vt') & a \\ t = \gamma(t' + \frac{Vx'}{c^2}) & b \end{matrix}} \quad (12)$

- what about y' and z' - ?

| 24. | 3

- Consider beam of light in the $+y'$ direction in S' .
Recall light clock from discussion 2.

In S light has y component of velocity $\sqrt{c^2 - v^2}$
(Due to (11)).

we have $y' = ct'$ and $y = \sqrt{c^2 - v^2} t = \frac{ct}{\gamma}$

But if our light clock is at $x' = 0$ in $S' \Rightarrow$
 $\Rightarrow t = \gamma t'$ (from Lorentz transformations) $\Rightarrow$

$\Rightarrow y' = ct' = c \frac{t}{\gamma} = y$

- Similarly, $z' = z$
- Therefore, The full set of Lorentz Transformation is:

$$\begin{aligned} t &= \gamma \left(t' + \frac{vx'}{c^2} \right) \\ x &= \gamma (x' + vt') \\ y &= y' \\ z &= z' \end{aligned}$$

for
relative
motion
in x direction.

Week 4, Lecture 4

Derivation of the Lorentz transformations

What about the Delta's?

We don't have to assume that the origins of S and S' are synchronized. Just like vectors have no preferred origins, and matrices implement linear transformations of vectors, Lorentz transformations implement linear transformations of **spacetime displacements**

Event A		Event B
(t_A, x_A)	$\Delta t = t_B - t_A, \Delta x = x_B - x_A$	(t_B, x_B)
(t'_A, x'_A)	$\Delta t' = t'_B - t'_A, \Delta x' = x'_B - x'_A$	(t'_B, x'_B)

Lorentz	$\Delta t = \gamma \left(\Delta t' + \frac{v}{c^2} \Delta x' \right)$
transformations:	$\Delta x = \gamma (\Delta x' + v \Delta t')$

Week 4, Lecture 4

Derivation of the Lorentz transformations

Summary

If frame S' moves with velocity v along the $+x$ direction with respect to frame S , and the origins of S and S' coincide at $t = t' = 0$, the relations between the coordinates (t, x, y, z) in S and (t', x', y', z') in S' are linear, and given by the Lorentz transformations:

$$\begin{aligned}t &= \gamma \left(t' + \frac{vx'}{c^2} \right) & y &= y' \\x &= \gamma(x' + vt') & z &= z'\end{aligned}$$

One Minute Paper: in which step(s) of the derivation did we invoke Postulate 1, the equivalence of the laws of physics in S and S' ?